

Insurance Logistic Regression Project Report

Logistic Regression — Predicting Life Insurance Purchase from Age

A simple, intuitive binary classification project that predicts whether a person will buy life insurance based on a single feature — their **age** — using `scikit-learn`'s Logistic Regression, and walks through the underlying sigmoid math by hand.

1. Project Objective

Build a logistic regression model to predict whether a person buys life insurance (**Yes/No**) using their age as the sole predictor, and demonstrate the model's internal mechanics by manually reproducing its predictions with the sigmoid function.

This is a **binary classification problem** — there are exactly two possible outcomes (bought insurance / did not).

2. Dataset

Attribute	Detail
File	<code>insurance_data.csv</code>
Observations	27 people
Feature (X)	<code>age</code>
Target (y)	<code>bought_insurance</code> (1 = Yes, 0 = No)
Class balance	14 bought (51.9%) / 13 did not (48.1%) — well balanced

Raw sample:

age	bought_insurance
22	0
25	0
47	1
52	0
46	1
56	1
18	0
60	1

3. Tools & Libraries

- **Python 3**
- `pandas` — data handling
- `matplotlib` — visualization
- `scikit-learn` — train/test split and `LogisticRegression`
- `math` — manual sigmoid calculation to verify the model's predictions by hand

4. Methodology

- 1. **Load the dataset** into a DataFrame.
- 2. **Explore visually** — a scatter plot of age vs. `bought_insurance` shows a clear pattern: younger people cluster at 0 (did not buy), older people cluster at 1 (bought), with a mixed zone in the middle.
- 3. **Train/test split** — 80% train / 20% test.
- 4. **Train the model** — `sklearn.linear_model.LogisticRegression` fit on `age` alone.
- 5. **Predict & evaluate** — generate class predictions and predicted probabilities on the test set, and score overall accuracy.
- 6. **Reproduce the model by hand** — extract the fitted coefficient (`m`) and intercept (`b`), plug them into the sigmoid function manually, and confirm the hand-calculated probabilities match `model.predict_proba()`.

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

X_train, X_test, y_train, y_test = train_test_split(
    df[['age']], df.bought_insurance, train_size=0.8
)

model = LogisticRegression()
model.fit(X_train, y_train)
y_predicted = model.predict(X_test)
```

5. Model Results

Term	Value
Coefficient (m)	0.1131
Intercept (b)	-4.1531
Fitted equation	<code>z = 0.1131 × age - 4.1531</code>

Model equation: `P(buys insurance) = sigmoid(0.1131 × age - 4.1531)`

Decision boundary: solving for `P = 0.5` gives an age of approximately **36.7 years** — the point at which the model's predicted outcome flips from "No" to "Yes".

6. Manual Verification with the Sigmoid Function

The notebook reproduces the model's predictions by hand to build intuition for how logistic regression actually works under the hood:

```
import math

def sigmoid(x):
    return 1 / (1 + math.exp(-x))

def prediction_function(age):
    z = 0.1131 * age - 4.1531
    return sigmoid(z)
```

Age	Hand-calculated P(buys insurance)	Interpretation
35	≈ 0.46	Below 0.5 → predicted No
43	≈ 0.72	Above 0.5 → predicted Yes

This confirms the model's internal logic matches `model.predict_proba()` exactly — a useful sanity check and a good

demonstration of understanding the math behind the library call, not just the API.

7. Model Evaluation

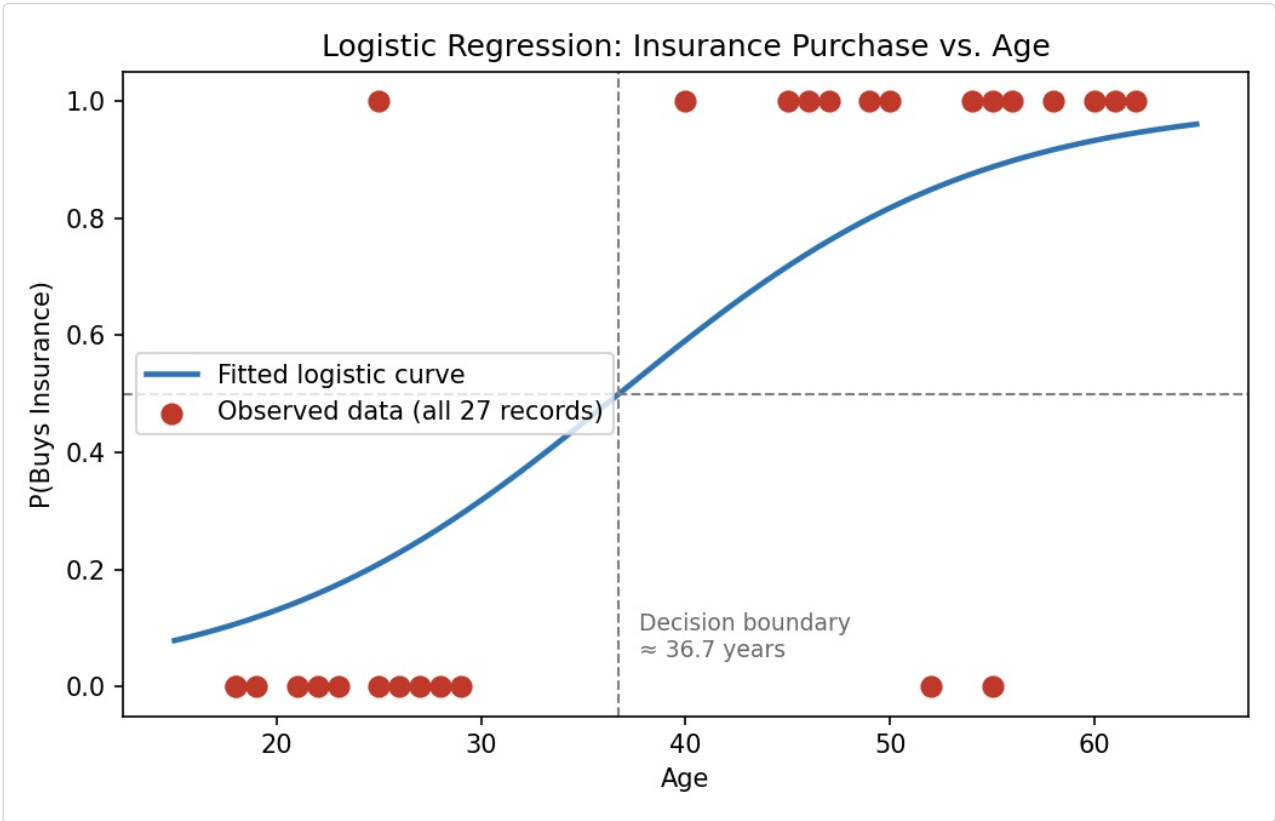
On the 20% held-out test set (6 people):

Age	Actual	Predicted	Predicted Probability (Yes)
62	Yes	Yes	0.946
61	Yes	Yes	0.940
29	No	No	0.294
28	No	No	0.272
26	No	No	0.229
22	No	No	0.159

Accuracy: 6 / 6 = 100% on this test split.

With only 27 total records and a 6-person test set, a perfect score is expected given how cleanly age separates the two classes here, rather than evidence the model would generalize this well on a larger, noisier population. This dataset is best treated as a clear, teaching-scale illustration of logistic regression mechanics rather than a production-ready model.

8. Visualization



The sigmoid curve shows the predicted probability of purchasing insurance rising smoothly from near 0 in the late teens to near 1 by the early 60s, with the steepest change — and greatest prediction uncertainty — right around the ~37-year decision boundary.

9. Key Insights

- **Age alone is a strong, intuitive predictor** of life insurance purchase in this dataset, consistent with real-world

intuition that older individuals are more likely to plan for end-of-life financial products.

- The **decision boundary of ~37 years** cleanly separates the observed data, with only a small "gray zone" (roughly ages 25–47) where outcomes are mixed.
- Manually reproducing the model's predictions with the sigmoid function confirms a solid understanding of what logistic regression is doing mathematically, beyond simply calling `.fit()` and `.predict()`.

10. Limitations & Next Steps

- **Very small dataset (27 records)** — the perfect test accuracy should not be over-interpreted; results would need validation on a much larger sample.
- **Single feature** — real insurance purchase decisions depend on income, dependents, health status, and other factors; a multivariate model would likely be more realistic.
- **No cross-validation** — a single train/test split can be sensitive to which records land in the test set; k-fold cross-validation would give a more robust accuracy estimate.
- The notebook's own exercise section suggests extending this workflow to a related HR dataset (employee retention vs. salary/department) as further practice — a natural next project applying the same logistic regression skills.

11. How to Run

```
pip install pandas matplotlib scikit-learn
jupyter notebook Insurance_logistic_regression_Answer.ipynb
```

12. Repository Contents

File	Description
<code>Insurance_logistic_regression_Answer.ipynb</code>	Main analysis notebook
<code>insurance_data.csv</code>	Source dataset
<code>insurance_logistic_curve.png</code>	Fitted sigmoid curve visualization
<code>README.md</code>	This report

13. Conclusion

This project builds a clean, interpretable logistic regression model that predicts life insurance purchase from age alone, achieving 100% accuracy on a small held-out test set and identifying an intuitive ~37-year decision boundary. Beyond fitting the model, the project reproduces its predictions by hand using the sigmoid function — reinforcing the mathematical foundation of logistic regression rather than treating it as a black box, which is a valuable habit for building toward more complex classification problems.

Tech stack: Python · pandas · matplotlib · scikit-learn